

Multi Asset Donchian - Strategy

Strategy Overview:

A long only donchian breakout strategy is run with a single, pre chosen parameters set across six crypto coins then combined into a combined portfolio. The main objective is to check whether diversification across multiple crypto assets can improve risk-adjusted performance compared with trading a single asset.

Data set:

BTCUSDT, ETHUSDT, LTCUSDT, BNBUSDT, XRPUSDT, and ETCUSDT

The data uses **hourly prices from August 2017 to May 2026**, covering multiple market regimes, including the 2021 bull market, the 2022 bear market, and the 2023–2025 recovery.

Strategy Rationale:

The strategy is based on a **long-only Donchian breakout system**. A buy signal is generated when price breaks above the previous Donchian upper channel. To reduce false breakouts, the model also applies trend, volume, and risk filters.

The core entry conditions are:

- Price closes above the upper Donchian channel.
- Price is above the 200-period EMA.
- ADX is above 22, confirming a trending market.
- Trading volume is at least 1.2 times the 20-period average volume.
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The exit rules include:

- Price falls below the lower Donchian channel.
- Price drops below an ATR-based stop-loss.
- Profit reaches 40%.
- Position is held for more than 21 days.
- Price falls below the 200-period EMA.

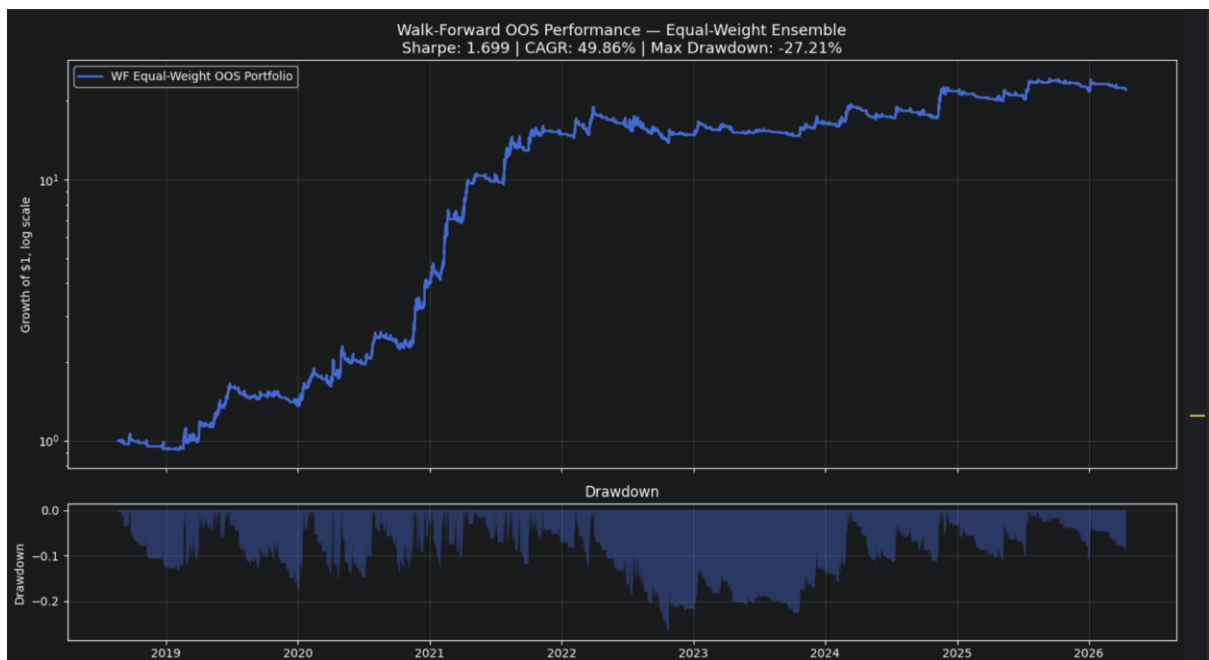
The strategy also uses **ATR-based position sizing**, so exposure is reduced during high-volatility periods and increased during calmer periods. Transaction costs are assumed to be **5 basis points per trade**, which is more realistic than using very low costs for hourly crypto trading.

Strategy performance:



Portfolio Construction:

Equal weight Portfolio: Each asset receives the same weight



Out of sample results:

Equal weight portfolio

Annual return: 49.86%

Sharpe ratio: 1.699

Profit factor: 1.099

Max Drawdown: -0.2721

The average pairwise correlation between the strategy returns is **0.357**, which shows that the assets are not perfectly correlated. This allows diversification to improve portfolio stability. However, since all assets are still from the crypto universe, the diversification benefit is limited.

Conclusion:

The results indicate that the Donchian breakout ensemble successfully captures trend-following behaviour in crypto markets. The high CAGR of **49.86%** shows strong return generation, while the Sharpe ratio of **1.699** suggests that the return was attractive relative to volatility.